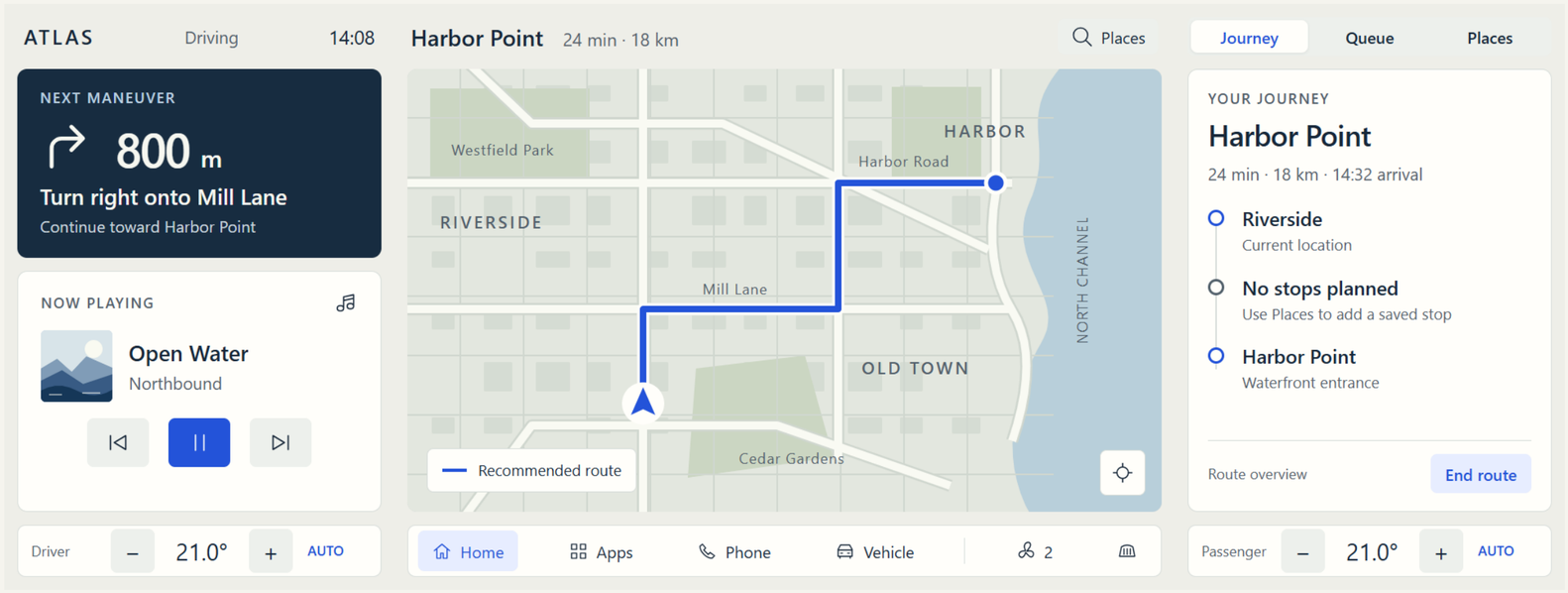
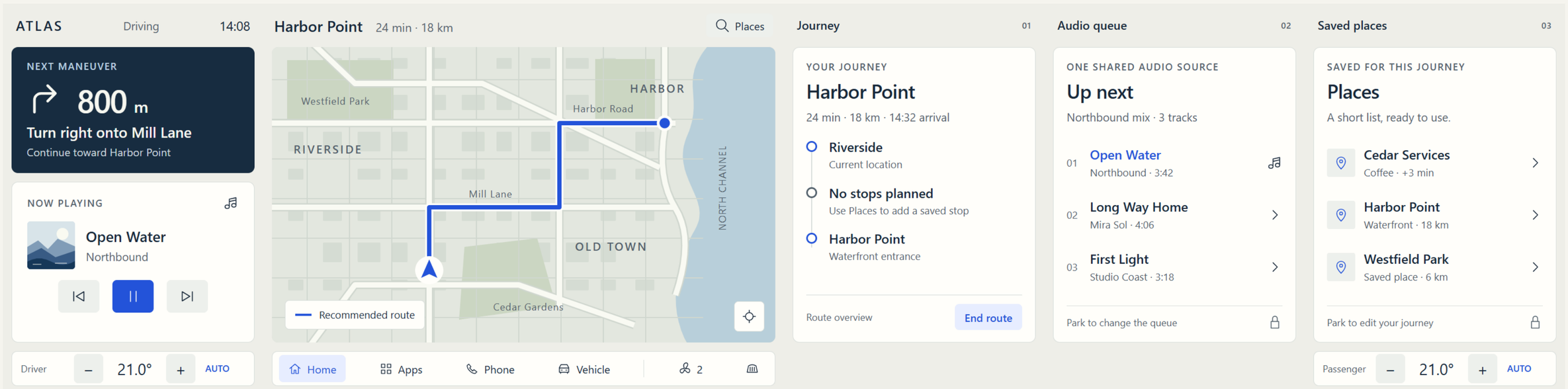
A stable core. An open task area.

1920 shows one shared task panel. 2880 unfolds three. The map and frequent driver controls do not grow or drift.

NORMAL / 1920 × 720



EXTENDED / 2880 × 720



Final Home screens at the same presentation scale. The larger display exposes the existing Queue and Places panels; it introduces no exclusive application.

The brief, translated into decisions

Deliver a concept explanation and visual documentation for one Home system at 1920 × 720 and 2880 × 720, with identical component dimensions and an economical implementation.

Standing	Decision
Required by the slides	Two display configurations; shared component pixel geometry; explain IA, screen zoning, layout optimization and cost-conscious reuse.
Erika's clarification	Widgets are acceptable. A growing map could work, but a fixed map is preferred. The additional area needs a purpose.
Chosen constraint	Keep the entire map viewport, widgets, headers, access controls and comfort groups fixed in width and height.
Working assumptions	Left-hand drive; added width to the right; separate instrument cluster; one journey and audio session; steering-wheel/physical volume and voice access assumed.
Scope	Home and a few illustrative states. No live routing, vehicle integration, passenger identification, full app suite or production validation.
Latest user direction	Start a fresh proposal. No previous layouts, visual styles, code assets or Figma kit assets are reused in this submission.

The two display sizes are hardware profiles, not a continuously fluid web layout. Only the canvas, task-panel count and passenger comfort anchor change.

The attached early implementation guidance contains superseded style and layout prescriptions. It is treated as project history; the slides and later user decisions govern this proposal. Earlier files are preserved, not deleted.

What the process needed to change

Earlier exploration generated useful questions, but visual detail repeatedly arrived before the adaptation logic. This pass closes the structural decisions first.

Step	Work completed	Decision produced
1. Reframe	Read the slide brief, existing guide and Erika's response. Separate requirements from historical ideas.	The fixed component pair is the deliverable; style is a separate decision.
2. Check evidence	Review OEM manuals, manufacturer publications, Apple guidance and standards scopes.	Borrow patterns, not unsupported claims of exact pixel reuse.
3. Define Home	Assign persistence, access and conditional content; declare one comparison scenario.	Navigation + compact playback + persistent controls; detail lives in reusable panels.
4. Compare mechanisms	Draw Distribute, Shared control bench and Unfold as paired block layouts.	Select Unfold for concurrency without moving the driver core.
5. Explore visuals	Render the same new composition in Paper, Tide and Signal treatments.	Select Paper + cobalt for the daylight concept.
6. Implement	One state store, geometry registry and shared renderer; build both hardware profiles.	No screen-specific application fork.
7. Verify	Inspect actual pixel bounds, overflow, state persistence, access and representative states.	Deliver measured geometry evidence and clearly bounded claims.

This is a designer-led concept evaluation, not a user study. No participant interviews, simulator sessions, measured glance tests or production cost estimates were conducted.

Research that changes the design

Industry precedents support the organization of functions. They do not prove our exact fixed-pixel solution.

Evidence	Transfer into ATLAS	Limit
Hyundai split screen [S1]	Existing functions can share a selectable secondary location.	Does not establish this two-size geometry.
BMW QuickSelect [S2]	Keep frequent activity access near the assumed driver.	No copied widget bar or unverified reach claim.
Volkswagen ID.7 [S3]	Separate persistent access/comfort from Home content.	Our controls are fixed groups, not stretching bars.
Apple vehicle integration [S4]	Separate the main task from adjacent functional regions.	CarPlay scaling/resizing is not borrowed.
Mercedes zero layer [S5]	Surface relevant content where it helps a task.	No AI recommendations or personalized prediction service.
NHTSA / ISO [S6-S8]	Treat distraction, dialogue and legibility as validation questions.	Public scopes are not certification; physical and driving tests remain.

A community kit or cockpit photograph can illustrate styling and grouping but cannot establish production adaptation rules. No screenshot was used as proof of ergonomic performance.

The original minimum-IA question becomes a product decision: this Home supports staying oriented, controlling audio and comfort, and preparing the current journey. It does not need a permanent weather, profile or connected-phone card.

One inventory, different visibility

The normal configuration has the same functions. The extended configuration changes how much of them can be seen at once.

Content / function	Home treatment	State rule
Next maneuver	Fixed driver-side guidance component	Active route; becomes a destination entry point when no route is active.
Map	Fixed central 928 × 544 viewport	Same viewport in both sizes; schematic asset, no live map service.
Now playing	Compact fixed driver-side widget	Playing/paused and track changes; controls remain near the driver.
Journey / Queue / Places	Three existing 448 × 544 panels	One tab-selected panel at 1920; all three visible at 2880.
Phone	Persistent access, no permanent card	Connection and illustrative call view use the same task area.
Apps / Vehicle / Home	Fixed access dock	Tools replace the first task panel; Home restores Journey.
Comfort	Driver and passenger temperature groups plus shared fan/defrost	Always available; passenger group follows the right edge.
Weather / profile / rich notifications	Excluded from this concept	Not needed for the defined tasks; no filler to justify width.

Scenario for comparison: an active journey to Harbor Point, one shared music queue and saved places. The map and all names/data are illustrative.

No predictive "context slot" is required. Places is explicitly saved content, and Journey accurately reports either no stops or the selected stop. This avoids a relevance engine and an arbitrary fallback card.

Three spatial hypotheses

Three width strategies. One component scale.

1920 and 2880 are drawn at the same 40% scale. Blue indicates additional visible instances.

01 / Distribute

Fixed components spread out; map shifts +480px. Lowest new-content cost, weakest use of added area.

NORMAL / 1920 × 720



EXTENDED / 2880 × 720



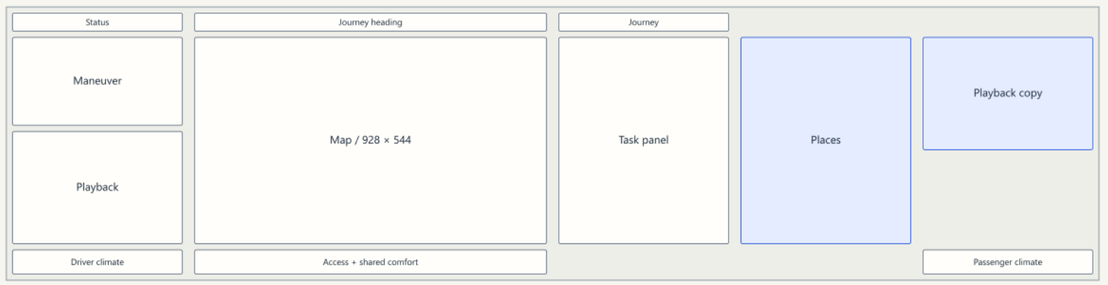
02 / Shared control bench

Keep the core and duplicate playback near the passenger. Useful access, but duplicated controls and state contention.

NORMAL / 1920 × 720



EXTENDED / 2880 × 720



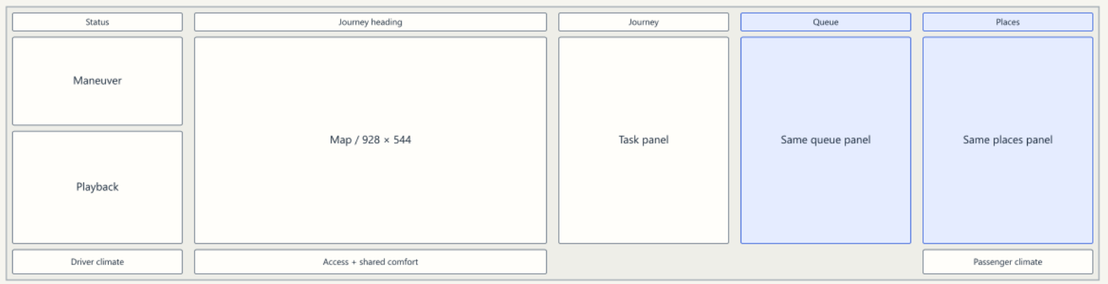
03 / Unfold existing panels — selected

Keep driver actions and map anchored. Reveal Queue and Places already available through normal-screen tabs.

NORMAL / 1920 × 720



EXTENDED / 2880 × 720



Each pair uses the same scale. The alternatives are structural studies, not independently implemented products. Their consequences are evaluated on the next page.

Why Unfold wins this brief

Criterion	Distribute	Shared control bench	Unfold · selected
Fixed component dimensions	Pass	Pass	Pass
Driver-relative consistency	Map and its access move +480px	Core stays fixed	Seven core components stay fixed
Use of +960px	Mostly separation; little new task value	Passenger-side shared access	Concurrent existing panels
Added behavior	Few new states	Duplicate playback controls need consistent feedback	Visibility rule and panel focus/recovery
Reuse	High; same instances	Shared components, more duplicate instances	Same panel views already available at 1920
Main risk	Space appears unearned; driver search shifts	Ambiguous control ownership / contention	More visible information; far-right reach unvalidated
Decision	Reject for weak width benefit	Keep as context-dependent alternative	Select for clear task benefit and bounded implementation

Unfold is still a visibility strategy. Its argument is not "more widgets are innovative"; it is that existing full task panels no longer compete for one slot. No exclusive weather, entertainment or passenger service is introduced to fill space.

A parked walkthrough illustrates the benefit: starting at Journey, choose a queued track, add the saved Cedar stop, then review Journey. Normal: 5 interface activations including three tab switches. Extended: 2 activations with Journey still visible. This is an interface-action count, not a measured safety, speed or usability improvement.

The narrower Home remains complete. Content was not removed from it to manufacture a premium-screen advantage.

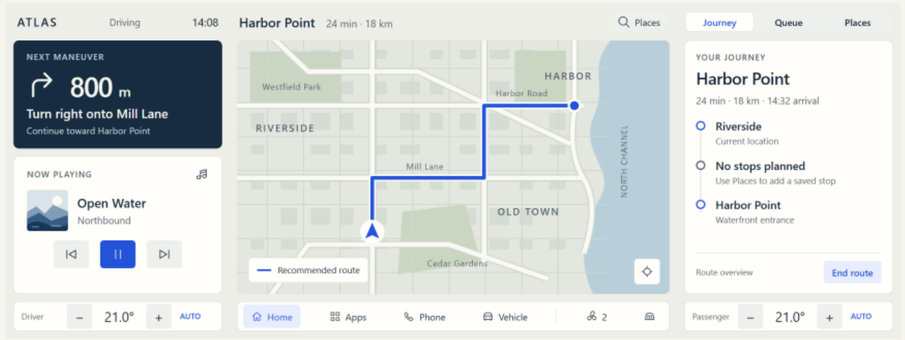
A new visual language, tested three ways

Visual exploration / three independent treatments

Identical content and geometry. No assets or style sheets inherited from the earlier concepts.

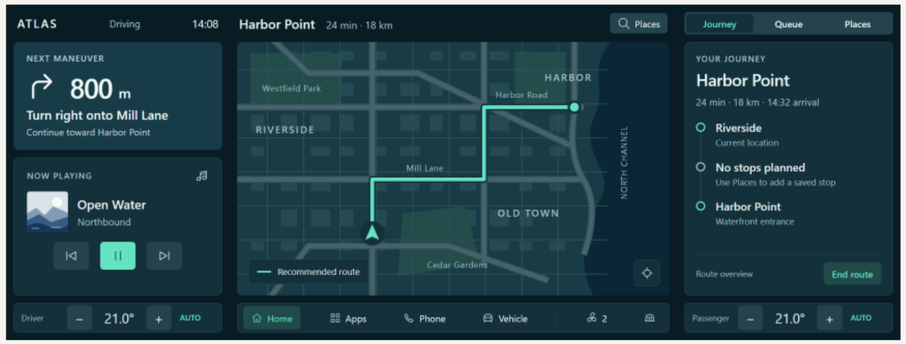
01 / Paper + cobalt — selected

Light surfaces, a dark maneuver anchor and precise blue route emphasis.



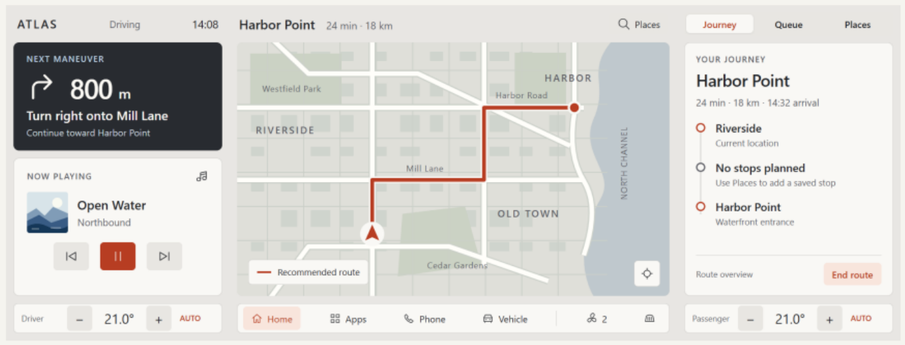
02 / Tide + mint

Dark instrument-like surface; useful night-direction candidate, but map tones need further tuning.



03 / Signal + vermilion

Neutral technical palette with a warm accent; route color may compete with warning semantics.



Why selected

Distinct information hierarchy; restrained decoration.

Best fit for the daylight review scenario.

Why not selected

Dark map labels require additional contrast work.

A visual study, not a validated production theme.

Why not selected

Warm route emphasis has a stronger alert association.

A visual study, not a validated production theme.

Paper + cobalt is selected. Tide + mint and Signal + vermilion remain visual studies using identical content and component dimensions; they are not validated day/night modes.

A layout contract, not two mockups

A 480px horizontal pitch makes +960px exactly two additional task locations. Modules keep 16px outer margins and 32px separation between pitch groups.

Component	Bounds	1920 position	2880 position
Status	448 × 48	16,16	16,16
Maneuver	448 × 232	16,80	16,80
Now playing	448 × 296	16,328	16,328
Map heading	928 × 48	496,16	496,16
Map viewport	928 × 544	496,80	496,80
Task panel	448 × 544	1456,80 (selected view)	1456 / 1936 / 2416,80
Task header	448 × 48	1456,16 (tabs)	1456 / 1936 / 2416,16 (titles)
Driver climate	448 × 64	16,640	16,640
Passenger climate	448 × 64	1456,640	2416,640
Access / shared comfort	928 × 64	496,640	496,640

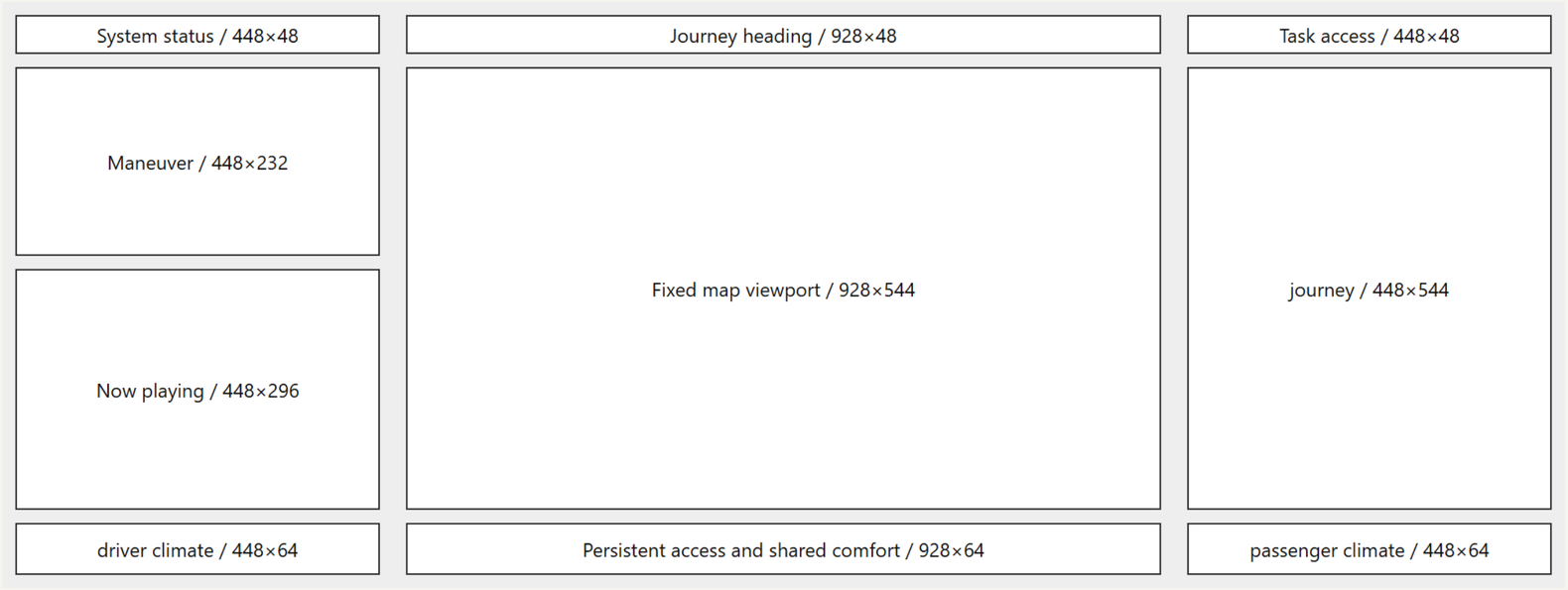
Horizontal equation: 480 driver region + 960 map region + (1 or 3) × 480 task regions = 1920 or 2880. Vertical equation: 16 margin + 48 header + 16 gap + 544 content + 16 gap + 64 controls + 16 margin = 720.

Task headers have two presentation states, tabs and title-only, at the same bounds. These are an explicit layout-dependent variation, not hidden resizing. The dock and status are fixed components; no full-width header or footer is stretched.

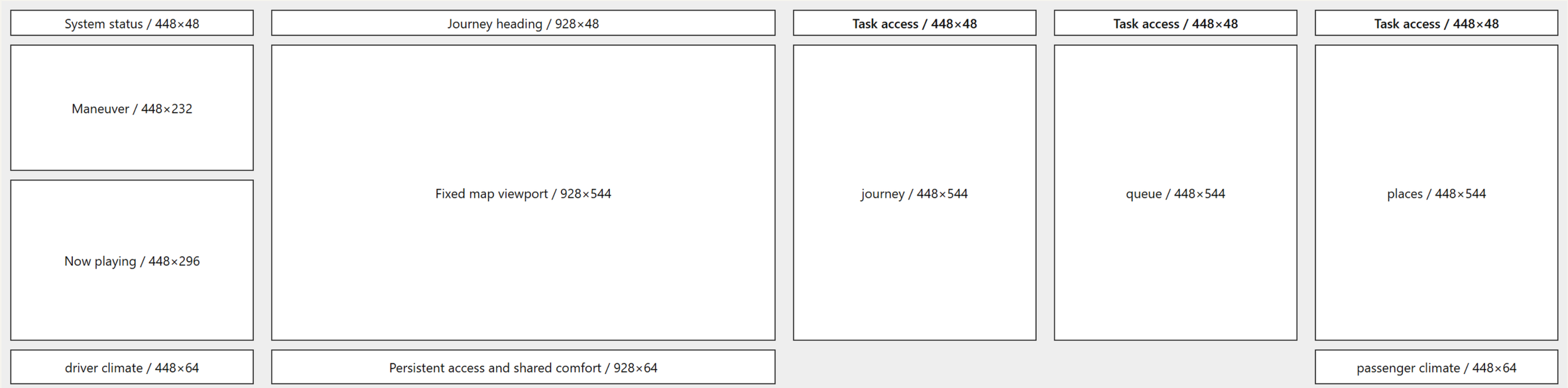
Passenger climate moves 960px with the passenger-side edge. That assumes rightward physical extension; it is not evidence of physical passenger reach. Empty lower space under the additional panels separates infrequent detail from persistent controls rather than adding new controls.

The selected structure at both sizes

NORMAL / 1920 × 720



EXTENDED / 2880 × 720



Normal and extended wireframes use the same presentation scale. These images come directly from the implemented component bounds, with decorative content suppressed.

States stay coherent across the pair

The prototype demonstrates the layout argument with a deliberately small set of interactions.

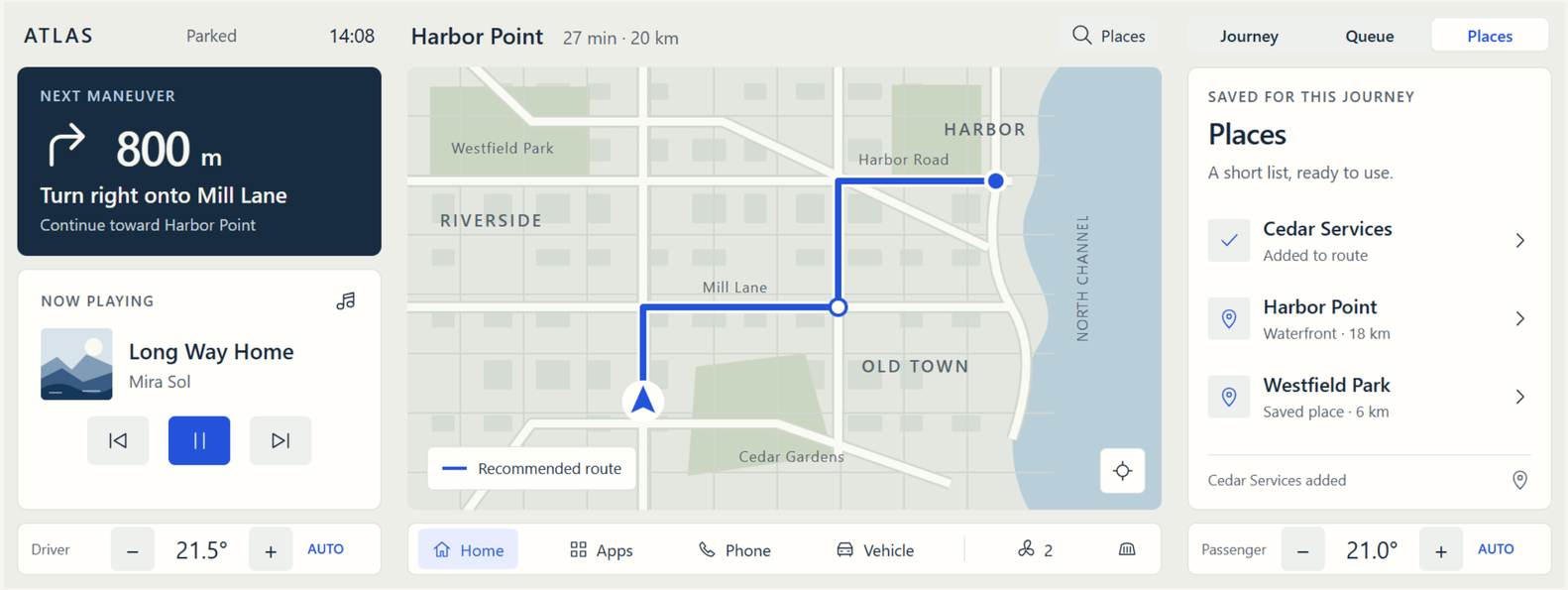
Situation	Response	Reuse / engineering consequence
Switch display profile	Keep journey, queue, playback, temperatures and connection state.	State store is independent from visible panel instances.
Switch normal-screen tabs	Replace only the secondary panel; map and driver controls stay visible.	Same renderer used when that panel is visible at 2880.
Choose a place / queued track	Allowed while parked; disabled during the on-route scenario.	One availability rule for both sizes; no passenger detection.
End route	Hide the route and update maneuver, map heading and Journey.	Geometry stays unchanged.
Open Phone / Vehicle / Apps	Replace the first task panel; Home recovers the default overview.	No separate full-screen application layouts in this concept.
Change playback / comfort	Immediate local state update in the existing control.	One audio session and shared vehicle settings.

Working controls include play/pause/skip, saved stop add/remove, route start/end, temperature changes, fan, defrost and illustrative phone connection/call states.

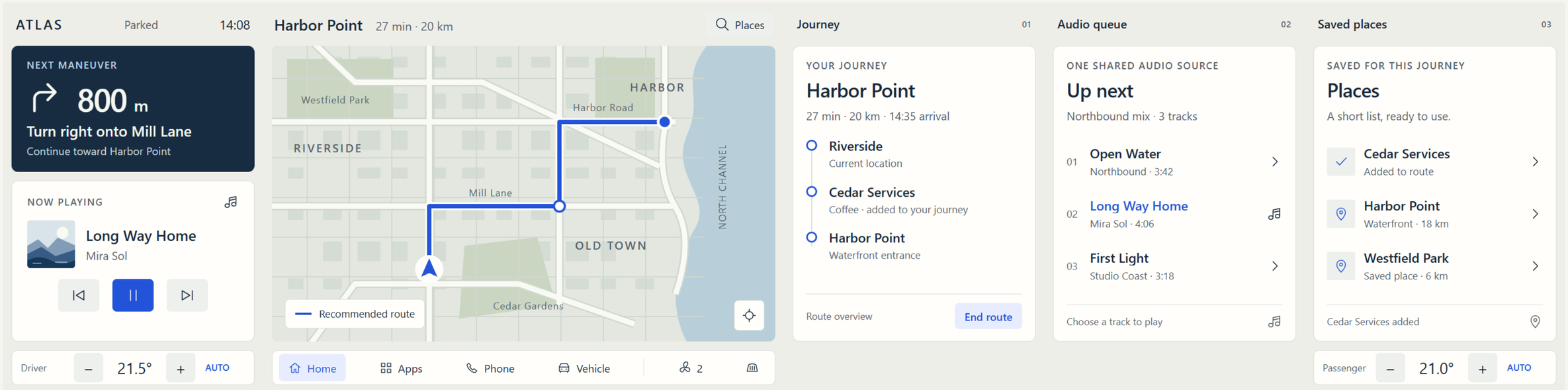
The map is schematic and static. Recenter confirms the already centered position; no live routing, map pan/zoom engine, real phone call, voice recognition or vehicle command is performed. Two saved-place entries show a preview notice; only Cedar Services changes the illustrative route. These are prototype limits, not completed integrations.

More space, the same underlying state

NORMAL / 1920 × 720



EXTENDED / 2880 × 720



Parked scenario: Cedar Services is added, Long Way Home is playing and driver temperature is 21.5°. Those same values persist when changing the display profile.

A useful extension has a cost

Reuse reduces variation; it does not eliminate the cost of showing additional views.

Area	Implementation in this prototype	Production implication
Top-level geometry	9 registered component types; 10 default instances at 1920, 14 at 2880.	Four extra instances: two task headers and two task panels; no new wide-only component type.
Content views	Journey, Queue, Places, Phone, Vehicle and Apps use the same fixed task frame.	Shared dimensions do not make these six bodies identical; each needs content/state tests.
Assets	One authored map SVG and one authored album illustration; shared icon definitions.	One map resource/render target in both profiles; real SDK performance still untested.
State	One store for route, media, comfort and connection.	A vehicle implementation needs service synchronization, error handling and lifecycle management.
Layout rules	Two hardware profiles; panel visibility; fixed anchors; right-anchored passenger climate; header presentation.	Configuration logic stays at the layout boundary. No per-widget width breakpoints.
Rendering	Simple full DOM render on state changes, appropriate to this demonstration.	Production should use targeted updates and retain renderer/service lifecycles.
QA	Shared component/state checks plus profile-specific bounds, focus and simultaneous-view checks.	Reused components still require testing in both compositions.

There are no estimated percentage savings. The defensible claim is a smaller variation surface than separate dashboards with resized map, compact/expanded widget bodies and exclusive passenger applications.

Additional mounted views still increase rendering work and simultaneous data presentation. The most economical design is not automatically the least populated one; the selected concept trades a small, explicit presentation cost for concurrent access.

What was verified, and what was not

The delivered prototype passes its structural and representative interaction checks. Human-factors performance remains a hypothesis.

Check	Evidence / result
Native dimensions	Screenshots and browser bounds: exactly 1920 × 720 and 2880 × 720.
Complete component reuse	Every mounted top-level component matches the shared geometry registry.
Anchor consistency	Seven core components preserve x, y, width and height. The first 1440 × 720 rendered pixels are identical in both default screenshots.
Content fit	No top-level component overflow in 48 representative content configurations. No overlapping components or out-of-canvas bounds in either default composition.
State persistence	Stop, selected track and driver temperature survive profile changes; playback and route toggles verified.
Availability / recovery	Queue and place editing disabled on route; enabled parked. Home, disconnected phone and route-ended states checked.
Visual review	Both final screens, three treatments, wireframes and representative state captures inspected.
Unvalidated	Physical reach, glare, PPI, glance behavior, driver workload, localization, vehicle performance and legal/standards compliance.
Task walkthrough	The automated parked sequence takes 5 activations at 1920 and 2 at 2880. This is a counted interaction sequence, not a user-performance result.

NHTSA guidance [S6] concerns distraction evaluation. ISO 15005 [S7] addresses dialogue, and ISO 15008 [S8] addresses specified visual presentation aspects. Their existence is not evidence this prototype meets them.

The largest unresolved product risk is whether two extra detail panels remain useful enough during a moving journey. Parked concurrency is demonstrated; actual driving benefit is not. If vehicle packaging puts the rightmost area out of useful view, this concept would need reassessment rather than a passenger label as justification.

Deliverables and final recommendation

Recommend ATLAS / Unfold as the submission concept: a stable driving core with concurrent shared task panels on the extended display.

Slide requirement	Delivered evidence
Concept explanation	This case study: IA, zone responsibilities, alternative arguments, chosen mechanism, engineering limits and evaluation.
Screen zoning and layout optimization	Three paired zoning hypotheses, selected wireframes, exact component coordinates and adaptation rules.
Home wireframes at both sizes	wireframe-1920.png and wireframe-2880.png.
Home prototypes at both sizes	prototype.html with both hardware profiles and shared interactive state.
Visual documentation	home-1920.png, home-2880.png, visual-exploration.png, zoning-alternatives.png and state captures.
Implementation/reuse evidence	prototype.js geometry registry, shared CSS and verification.json.
Process and research record	case-study.md, case-study.html, this PDF and linked primary sources.

The final decision is provisional only in the product-development sense: it is the recommended answer to the exercise, not a certified production HMI. The required concept and two-size prototype deliverables are complete.

Next in a real program: confirm vehicle packaging and hardware controls, evaluate representative tasks in a cockpit/simulator, then integrate live vehicle and infotainment services. Those are external validation steps rather than missing work needed to explain this thought-project solution.

Sources and evidence boundaries

S1 / Hyundai - Basic operation and split screen

A selectable secondary pane can show different existing functions. This supports reusable panel access, not identical pixel dimensions across models.

S2 / BMW - Operating System 8.5 / QuickSelect

BMW describes a driver-side widget bar and direct function access beside a large selected view. This supports protecting frequent access near the driver.

S3 / Volkswagen - ID.7 interior concept and technologies

Persistent access and comfort layers surround variable Home tiles. Their responsibilities can be separated without copying full-width bars.

S4 / Apple - Optimize CarPlay for vehicle systems (WWDC23)

CarPlay occupies an allocated view area and can coexist with native UI. Dynamic resizing is available; that mechanism is not compatible with this project's strict fixed map.

S5 / Mercedes-Benz - MBUX reaches a new level

Relevant functions can be surfaced at the top level. ATLAS uses explicit saved data and deterministic states, not a new prediction system.

S6 / NHTSA - Final visual-manual distraction guidelines (2013)

Voluntary guidelines assess visual-manual secondary tasks. A screen image or pixel-level test cannot demonstrate distraction compliance.

S7 / ISO 15005:2017 - Dialogue management

Public scope concerns driver-system dialogue and verification conditions. Used to frame interaction consistency; the full standard was not audited.

S8 / ISO 15008:2017 - In-vehicle visual presentation

Public scope covers dynamic visual information and legibility; maps/topographic representations are explicitly excluded from its scope. No blanket map-compliance claim is made.

Accessed 26 September 2026. OEM examples support patterns; none is presented as proof of this exact geometry. ISO references use public scope descriptions, not a paid full-standard audit.